Evaluating AI Agents

Adding Router & Skill Evaluations





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Talking Head

In this lesson:

- Techniques for running evaluations (evals):
  - LLM-as-a-Judge
  - Code-based evals
  - Human Annotations
- Router evals
  - Function calling, parameter extraction, function generation
- Skill evals

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Techniques for Running Evals

Code-Based Evals
Running code to compare outputs to expected outputs, run calculations on outputs, etc.

LLM-as-a-Judge Evals
Using a separate LLM to judge the outputs of your application.

Human Annotations
Use human labelers or user feedback to evaluate your application outputs

Code-based Evals

Code run on the outputs of your application

Examples of code-based evaluators:

- Matches regex
  - E.g. response only contains numbers
- JSON parseable
- Contains keyword(s)
  - E.g. response contains competitor name

Code-based Evals

Can also be used to compare your application's outputs to **expected outputs**

Can be compared using:

- Direct match
- Cosine similarity / cosine distance

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LLM-as-a-Judge

PROMPT

QUERY I want to go...
CONTEXT User booking history, hotel inventory...
INSTRUCTIONS You are a friendly travel assistant...



APP IN PRODUCTION

Output

{output} Hi, I'm happy to help you plan...

EVAL PROMPT TEMPLATE

<<template: hallucination>>
Is the {output} using the {context} to answer the {query}

EVALUATOR LLM

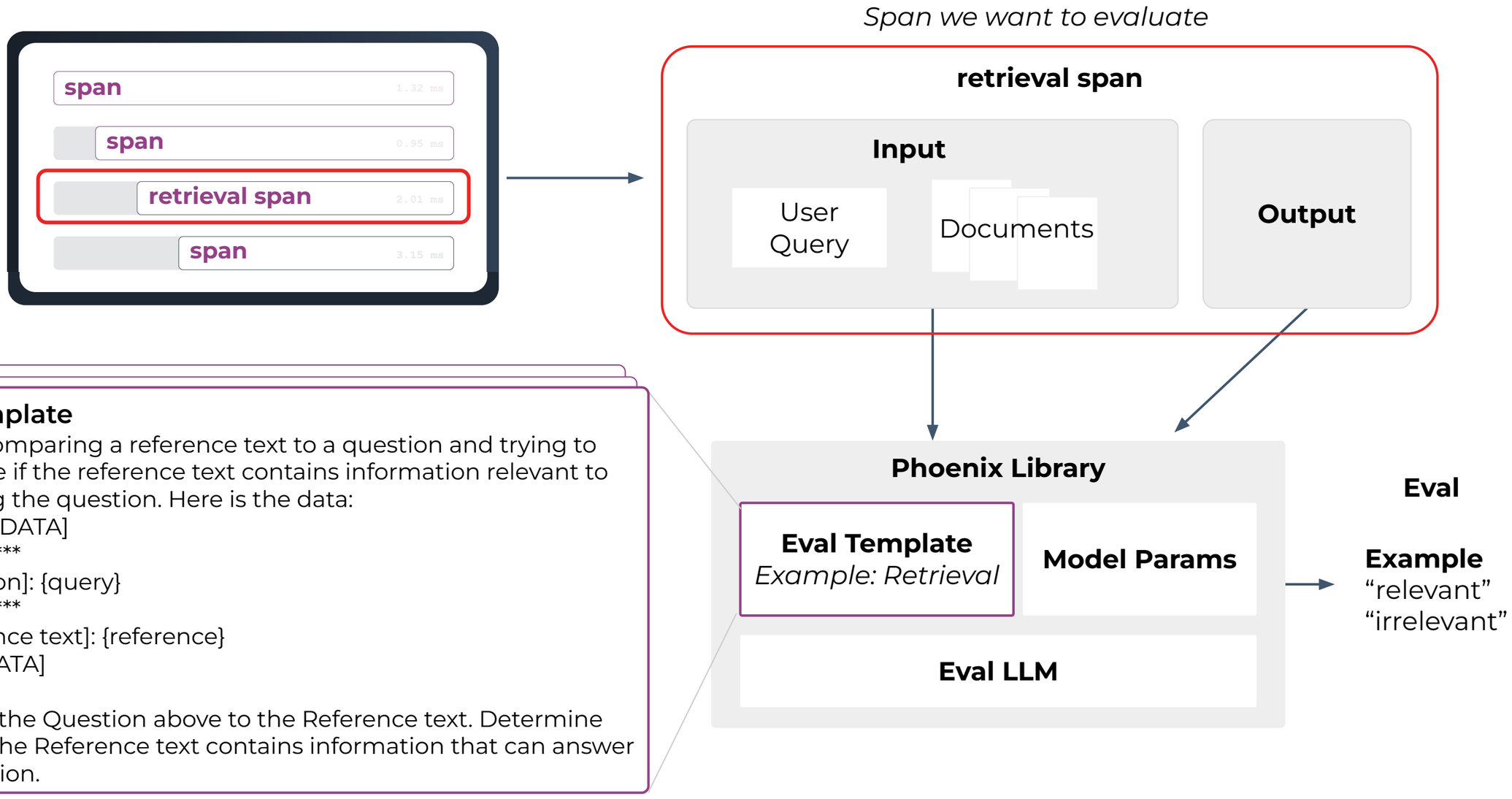
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EVAL

Did it hallucinate?

EVAL *No*

LLM-as-a-Judge Example



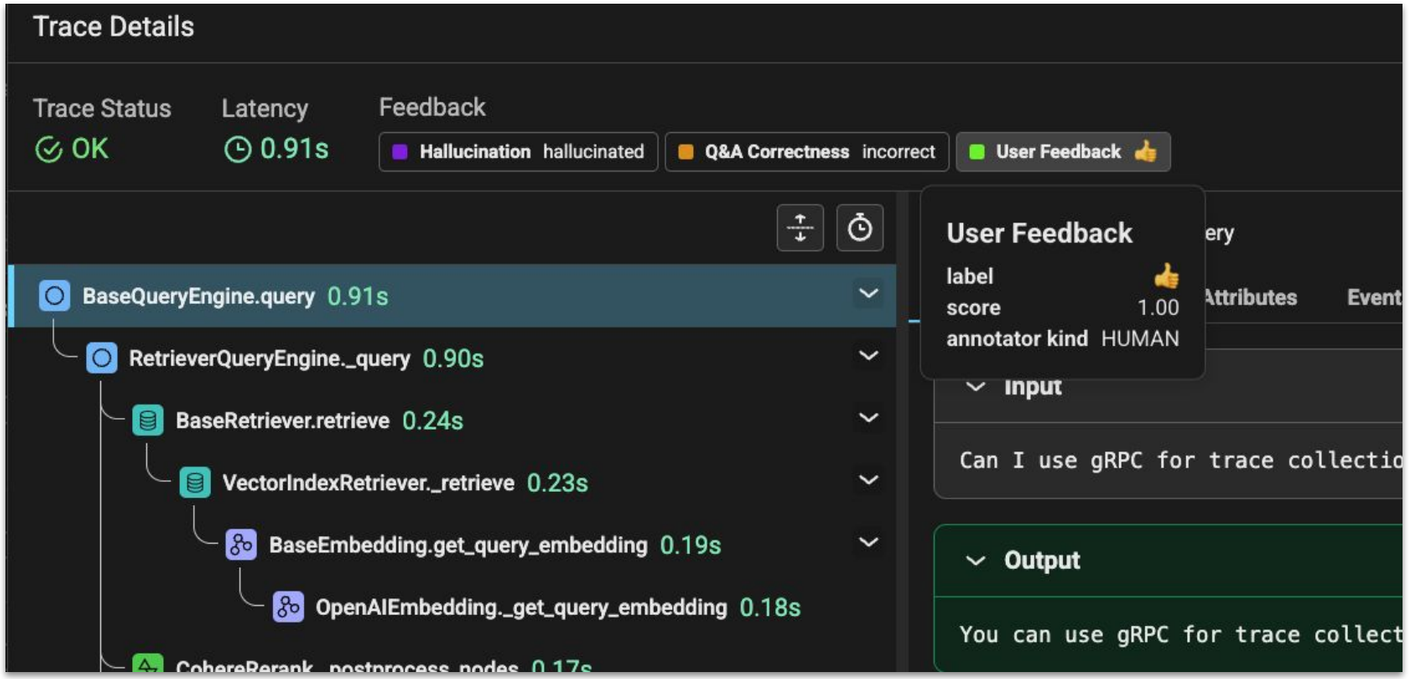
Important considerations when using LLM Judges

1. Only the best models align closely with human judgement
2. LLM-as-a-Judge will never be a 100% correct technique
3. Tuning your LLM judge prompt can help close this gap
4. Always use discrete classification labels, never an undefined score
    a. “Incorrect” vs “correct”, not 1%-100% accuracy

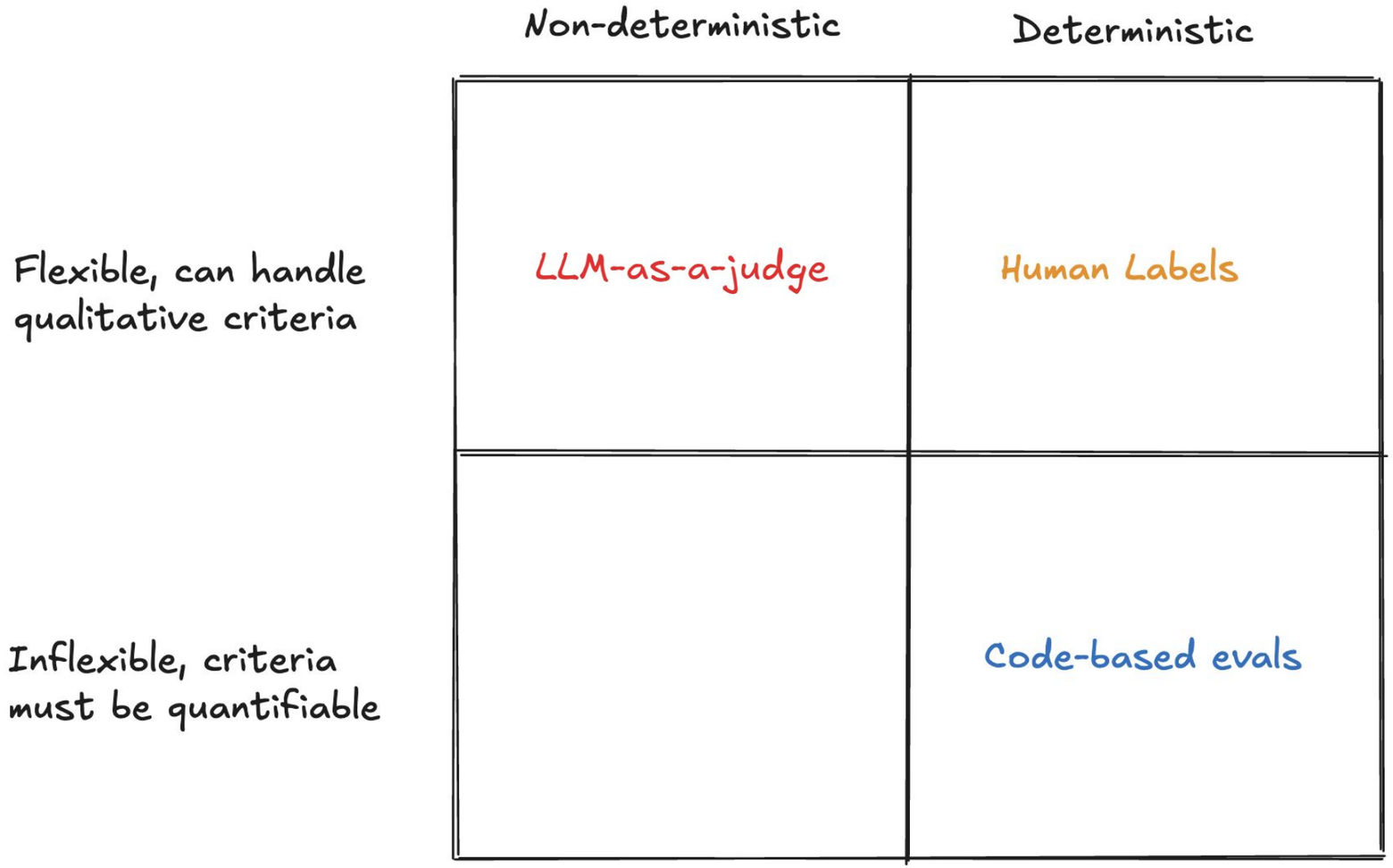
Annotations

Add human evaluation labels to your traces:

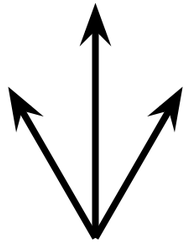
- Using annotation queues and human labelers
- Using 👍 👎 responses from users



When to use each Technique



Choosing what to evaluate



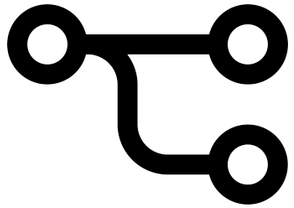
The Router

Function choice and parameter extraction



The Skills / Functions

Can use standard LLM evaluations



The Path

The most challenging to evaluate scalably

Evaluating a Router

Routers are usually evaluated in two ways:

1. **Function calling choice** - did the router choose the right function to call?
2. **Parameter extraction** - did the router extract the right function parameters from the question?

Evaluating a Router

User: Hi, can you help check on the status of my order? #1234

Agent: Definitely!

- `{tool_call: "order_status_check(order_number=1234)"}`
- `{role: "tool", content: "status=shipped"}`

Your order has been shipped!

User: When will it arrive?

Agent: Let me check.

- `{tool_call: "shipping_status_check(shipping_tracking_id=1234)"}`
- `{role: "tool", content: "tracking id not found"}`

Evaluating a Router using LLM-as-a-Judge

```
TOOL_CALLING_PROMPT_TEMPLATE = """
You are an evaluation assistant evaluating questions and tool calls to determine whether the tool called would
answer the question. The tool calls have been generated by a separate agent, and chosen from the list of tools
provided below. It is your job to decide whether that agent chose the right tool to call.

    [BEGIN DATA]
    ************
    [Question]: {question}
    ************
    [Tool Called]: {tool_call}
    [END DATA]

Your response must be single word, either "correct" or "incorrect", and should not contain any text or
characters aside from that word.
"incorrect" means that the chosen tool would not answer the question, the tool includes information that is
not presented in the question, or that the tool signature includes parameter values that don't match the
formats specified in the tool signatures below.
"correct" means the correct tool call was chosen, the correct parameters were extracted from the question, the
tool call generated is runnable and correct, and that no outside information not present in the question was
used in the generated question.

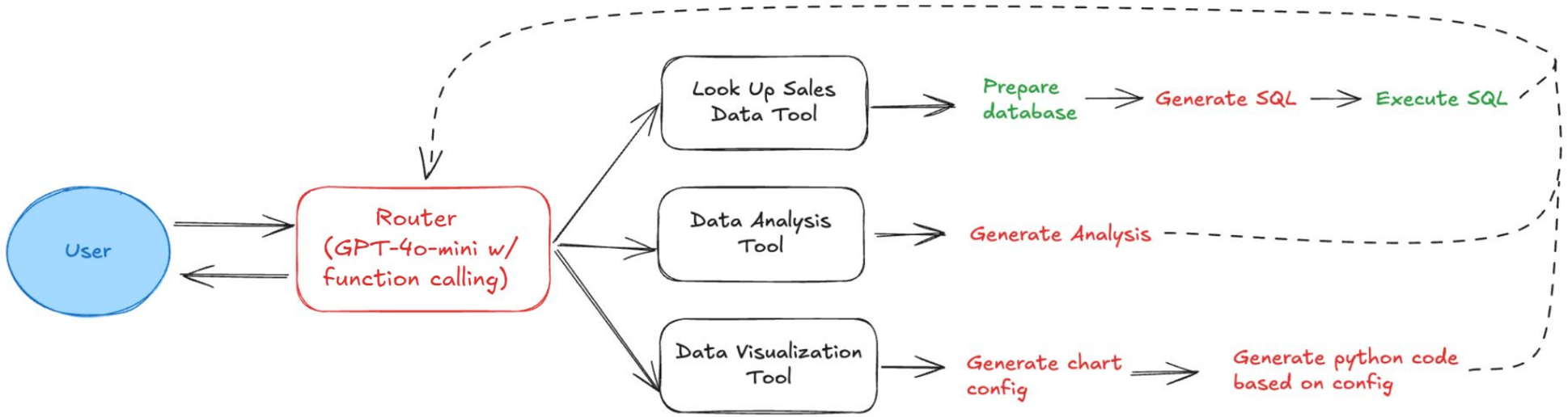
    [Tool Definitions]: {tool_definitions}
"""
```

Evaluating Skills

Skills can be evaluated using standard LLM or code-based evals:

- Relevance
- Hallucination
- Question and answer correctness
- Generated code readability
- Summarization
- Regex
- JSON parseable
- ...

Evaluating Skills - Your Example Agent



Tool #1:

Database Lookup

?

Tool #2:

Data Analysis

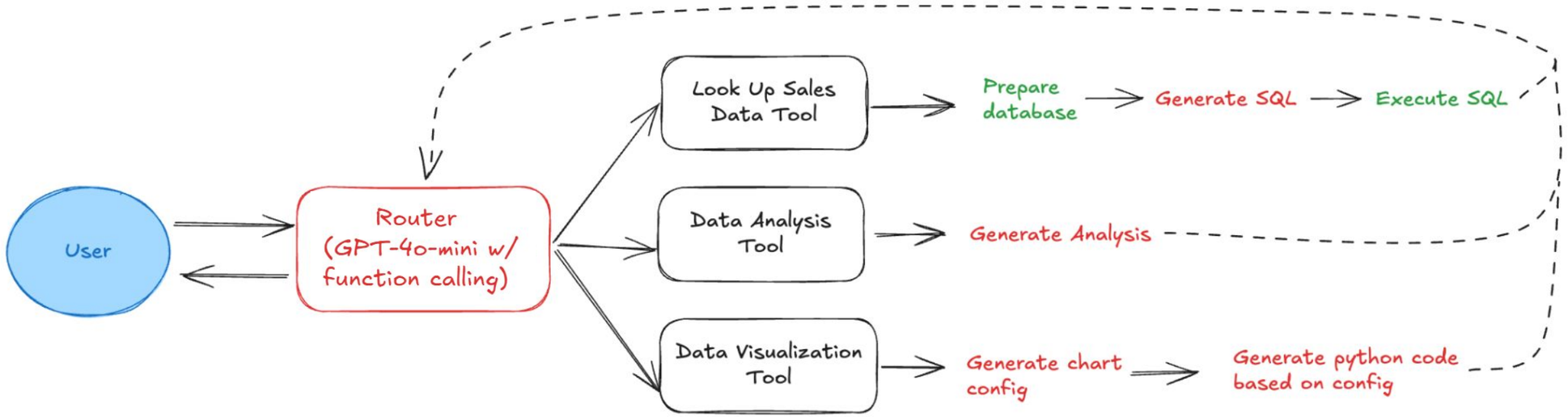
?

Tool #3:

Data Visualization Code Gen

?

Evaluating Skills - Your Example Agent



Tool #1:

Database Lookup

- SQL gen correctness
- SQL gen correctness

Tool #2:

Data Analysis

- Analysis Clarity
- Entity correctness

Tool #3:

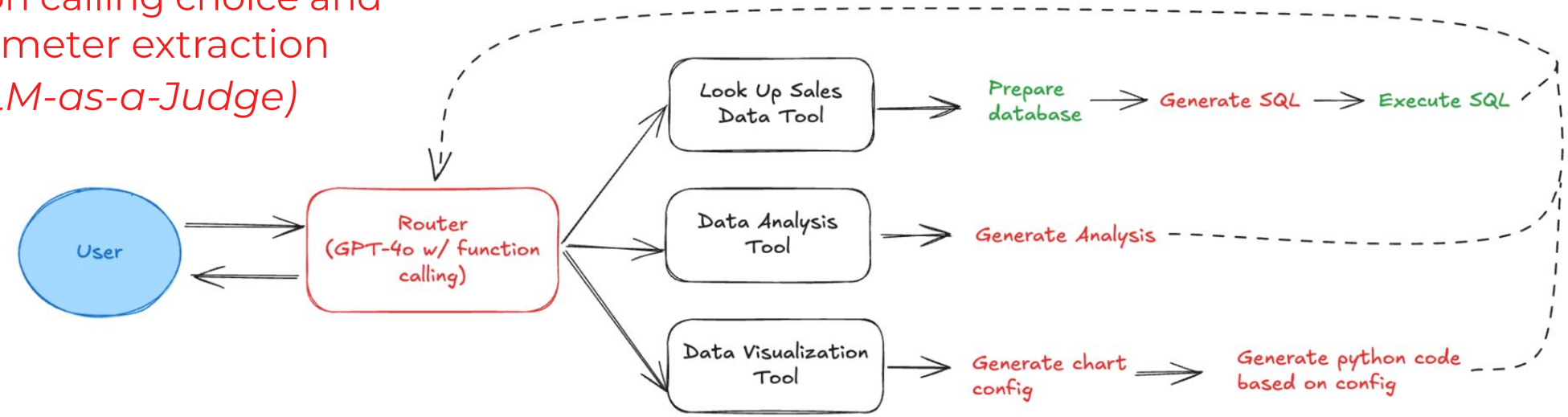
Data Visualization Code Gen

- Code is runnable

- Using LLM-as-a-Judge
- Using code-based evals

Router:

Function calling choice and parameter extraction *(LLM-as-a-Judge)*



Tool #1:

Database Lookup

SQL gen correctness *(code-based)*

Tool #2:

Data Analysis

- Analysis Clarity
- Entity Correctness

(LLM-as-a-Judge)

Tool #3:

Data Visualization Code Gen

Code is runnable *(code-based)*

What you've learned:

1. Three techniques to run Agent Evals: LLM-as-a-Judge, code-based evals, and human annotations
2. Common types of evaluations to run using these techniques
3. Which pieces of your agent to evaluate